



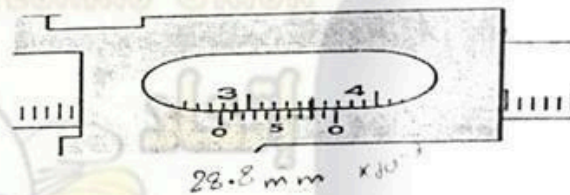
Midterm Exam Phy-111(C)

اسم الطالب: ()
المادة: فيزياء (111) موعده الاختبار: (11 - 12)

الاسم بالعربية: ()

Q 1: Fill the blank with suitable answer : (13.5 marks)

- 1) If we turned a closed micrometer (9) turns and (1/5) of a turn, then the micrometer reading is (7.5 mm).



2) $L = (0.028 \text{ m}) \pm$

$\Delta L = (0.005 \text{ m})$

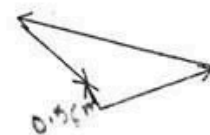


$D = 2.12$

$r = 1.06 \text{ mm}$

- 3) If the diameter for a ball was measured using a micrometer as shown then: $\Delta A = (1.41 \times 10^{-6} \text{ m}^2)$.

- 4) If the scale used to draw this vector diagram for three forces was (2mm = 3 N), then the magnitude of the net force is (7.5)



- 5) The instrument used to take this reading (11.35 cm) is (Vernier caliper, Micrometer, Ruler).

(vernier caliper)

6) Using significant rules to find

$$37600/4 = (0.9 \times 10^4)$$

7) In gravitational potential energy experiment, before start move toward down from ($h_1=70\text{cm}$) the mass was ($m=20\text{g}$) and ($M=200\text{g}$), then the kinetic energy when (m) reach ($h=40\text{cm}$) is: (0.058 joule).

8) For the above question the final velocity (V_f) when m reach ground is: (1.46 m/s).

9) Using significant rules to find , $6.299+3.78=(10.08)$.

• Q2 : Circle the correct answer: (10.5 marks)

1) If a student used a ruler with a maximum range (20 cm) to measure the length of a table which has the length (1.7 m), then the uncertainty in length (ΔL) is:

a) 3mm

b) 4.5mm

☒ c) 8.5mm

d) 6mm

2) If the relation between A & B given as ($C=A-B$), if $A=5.28$ and $B=1.8$, then ($C \pm \Delta C$) is:

(Using significant rules)

a) 3.48 ± 0.05

b) 3.5 ± 0.1

☒ c) 3.5 ± 0.06

d) 3.48 ± 0.055

3) F_1, F_2, F_3 are three forces in equilibrium, if you know that ($F_{3x}=1.32\text{ N}$, $F_{3y}=7.5\text{ N}$), the direction of resultant between (F_1 & F_2) according to x-positive axis is :

a) 100

b) 80

c) 280

☒ d) 260

4) In Newton's second law experiment, if we got this graph,

then the constant hanged mass in the experiment is:

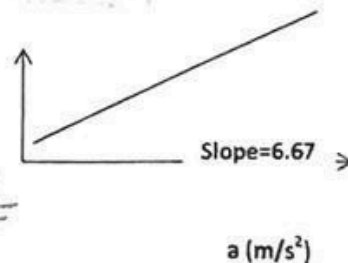
☒ a) 15g

b) 25g

c) 60g

d) 80g

$1/Mw+mh(\text{kg}^{-1})$



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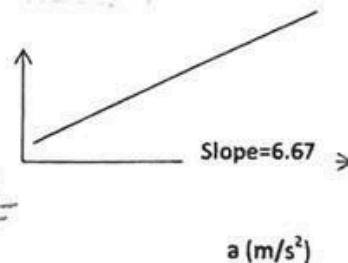
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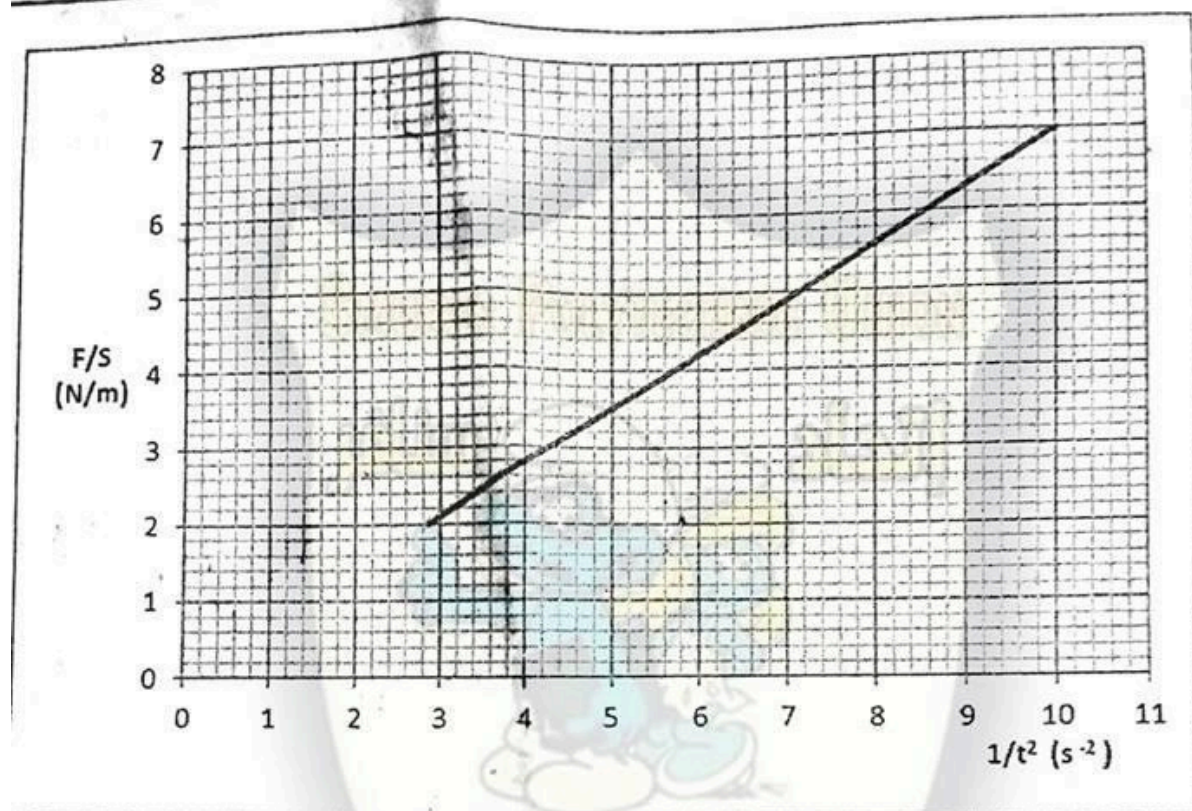
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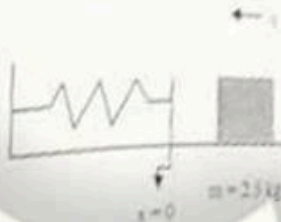


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13. Suppose that the work done by an engine is given by $W = 7t^2 + 40t + 100$, where the units are SI. The power (in watt) developed by this engine at $t = 2$ s is:

- a) 68 b) 82 c) 96 d) 54

14. A mass $m = 2.5$ kg is sliding along a frictionless table with initial speed v as shown in the figure. It strikes a spring that has a force constant $k = 1000$ N/m and compresses it from its equilibrium position a distance 5 cm ($x_f = -5$ cm). The initial speed v of the block was (in m/s):



- a) 1.22 b) 1.41 c) 0.71 d) 1

15. A sphere of mass m slides on a frictionless path ABC. If the sphere starts from rest at point A, the speed of the sphere (in m/s) at point C is:



- a) 24.5 b) 17.3 c) 14.1 d) 31.6

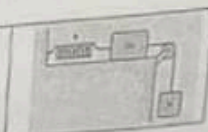
16. The potential energy function for a system is given by $U(x) = -x^3 - 2x^2 + 3x$. The magnitude of the force (in N) at $x = 3$ m is:

- a) 92 N b) 17 N c) 36 N d) 61 N

17. A particle ($m = 4$ kg) moves along the x axis. Its position varies with time according to $x(t) = t^2 + 2t$, where x is in meters and t is in seconds. The kinetic energy of the particle at time $t = 3$ s:

- a) 200 J b) 128 J c) 72 J d) 32 J

18. The blocks shown are released from rest with the spring unstretched. The pulley and the horizontal surface are frictionless. If $k = 150$ N/m and $M = 1$ kg, what is the maximum extension of the spring?



- a) 13.3 cm b) 26.7 cm c) 20 cm d) 6.7 cm

6. A spring has compressed a distance $x = 0.1$ m. If the spring constant $k = 200$ N/m, calculate the potential energy stored in the spring?

- a) 1 J b) 2.25 J c) 4 J d) 0.25 J

7. A force F ($F = 40$ N), parallel to the incline, is required to push a certain object at constant velocity up a frictionless incline that is 30° above the horizontal. The mass of the object is:

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- a) 6.4 kg b) 8.8 kg c) 10.4 kg d) 8.0 kg

8. 2000 kg car - moving at an initial speed of 25 m/s along a horizontal road - can brake to stop in a distance 80 m. The coefficient of kinetic friction between the tires and the road is:

- a) 0.26 b) 0.39 c) 0.52 d) 0.31

9. When an object is moving in a circle at constant speed, its acceleration is

- a. constantly increasing. b. constant in direction.
c. zero. d. constant in magnitude.

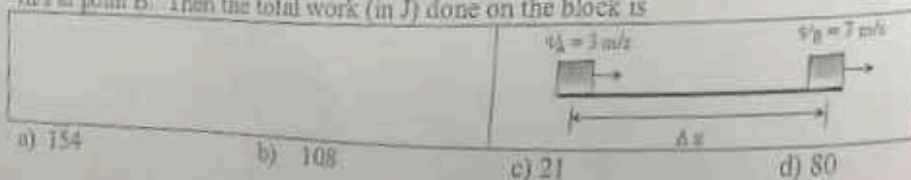
10. A racing car rounds a curve at a speed of 30 m/s. If the force needed to provide the centripetal acceleration is equal to the weight of the car, the radius of the curve is:

- a) 90 m b) 160 m c) 250 m d) 40 m

11. The coefficient of static friction between a 6 kg-block and horizontal surface is 0.4. What is the maximum horizontal force that can be applied to the block before it moves?

- a) 32 N b) 20 N c) 24 N d) 28 N

12. A block of mass = 4 kg has a speed of 3 m/s initially (at point A) and a speed of 7 m/s at point B. Then the total work (in J) done on the block is



- a) 154 b) 108 c) 21 d) 80